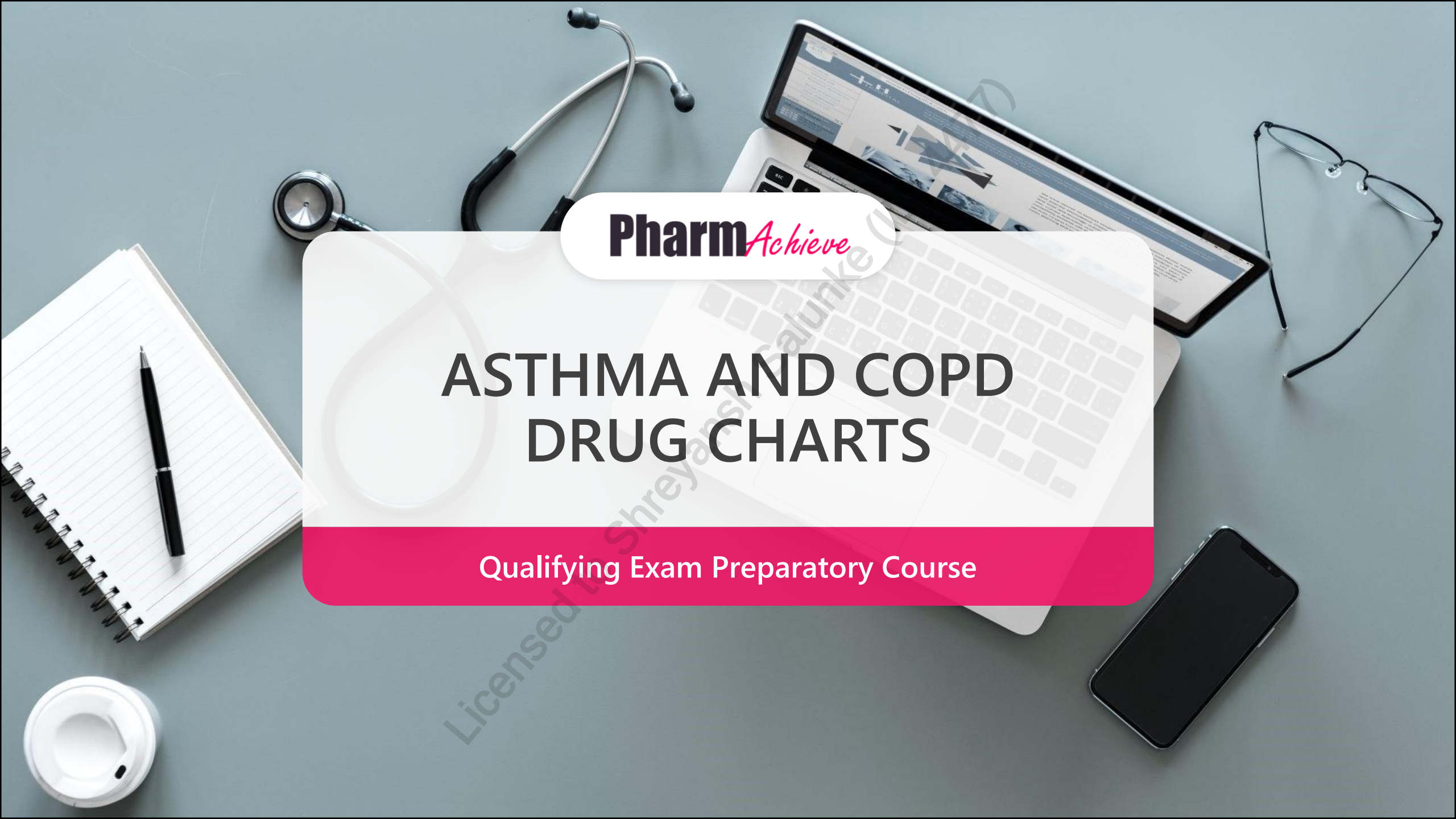




Pharm*Achieve*

ASTHMA AND COPD DRUG CHARTS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

COX Cyclooxygenase 1 Inhibitor

D/I Drug Interactions

DPI Dry Powder Inhaler

ICS Inhaled Corticosteroid

IV Intravenous

LABA Long-Acting Beta-2 Agonist

LAMA Long-Acting Muscarinic Antagonist

LTRA Leukotriene Receptor Antagonist

MDI Metered Dose Inhaler

NSAID Non-Steroidal Anti-Inflammatory Drug

SABA Short-Acting Beta-2 Agonist

SAMA Short-Acting Muscarinic Antagonist

SC Subcutaneous

SHORT-ACTING BETA-2 AGONISTS (SABA)

Drugs	MOA	Safety	D/I
• Salbutamol • Terbutaline	Causes bronchodilation by relaxing airway smooth muscle	• Tachycardia, • Nervousness (for 1-2 hours after high dose use) • Skeletal muscle tremors • Hypokalemia and hyperglycemia • Palpitations	• Steroids • Sympathomimetics • Beta-Blockers • Diuretics • Digoxin

SHORT-ACTING MUSCARINIC ANTAGONISTS (SAMA)

Drugs	MOA	Safety	D/I
Ipratropium	Inhibits acetylcholine-induced bronchoconstriction	- Dry mouth, difficulty urinating - Metallic taste - Mydriasis and glaucoma (if released into eye) - Possible increase in cardiovascular events	Other anticholinergic agents

LONG-ACTING BETA-2 AGONISTS (LABA)

Drugs	MOA	Safety	D/I
• Formoterol fumarate dihydrate • Salmeterol • Indacaterol • Olodaterol • Vilanterol	Causes bronchodilation by relaxing airway smooth muscle	• Tachycardia • Nervousness (for 1-2 hours after high dose use) • Skeletal muscle tremors • Hypokalemia and hyperglycemia • Palpitations	• Beta-Blockers • Steroids • Sympathomimetics • Diuretics • Digoxin

LONG-ACTING MUSCARINIC ANTAGONISTS (LAMA)

Drugs	MOA	Safety	D/I
• Tiotropium • Aclidinium • Umeclidinium • Glycopyrronium	Inhibits acetylcholine-induced bronchoconstriction	• Dry mouth, difficulty urinating • Metallic taste • Mydriasis and glaucoma (if released into eye) • Possible increase in cardiovascular events	• Other anticholinergic agents

INHALED CORTICOSTEROIDS (ICS)

Drugs	MOA	Safety	D/I
• Beclomethasone • Budesonide • Ciclesonide • Fluticasone propionate • Fluticasone furoate • Mometasone furoate	↓ inflammation	• Sore mouth and sore throat, dysphonia • Oral thrush (use spacer/rinse)	• Strong CYP 3A4 inhibitors (exception beclomethasone)

LEUKOTRIENE RECEPTOR ANTAGONIST (LTRA)

Drugs	MOA	Safety	D/I
• Montelukast (oral)	Reduces broncho-constriction and inflammation	• Headache • Abdominal pain • Flu-like symptoms • Nightmares • Neuropsychiatric symptoms	• Carbamazepine, rifampin, phenobarbital, phenytoin

METHYLYXANTHINES

Drugs	MOA	Safety	D/I
• Theophylline (oral) • Aminophylline (IV)	Relaxation of bronchial smooth muscle	• Nausea & vomiting • Abdominal cramps • Headaches • Palpitations • CNS stimulation • Contraindicated in coronary artery disease • Narrow therapeutic index, therapeutic drug monitoring required	• Estrogens • Phenytoin, carbamazepine, rifampin • Quinolones • Macrolides • Not an exhaustive list (CYP1A2, CYP3A4 substrate)

BIOLOGICS

Drugs	MOA	Safety	D/I
• Omalizumab (SC)	Binds to IgE receptors to reduce allergic inflammatory cascade	• Injection site reactions • Viral infections • Upper respiratory tract infections • Headache	• Low interaction potential
• Mepolizumab (SC) • Benralizumab (SC) • Reslizumab (IV)	Reduces the production and survival of eosinophils by inhibiting IL-5 activity	• Headache • Injection site reactions • Back pain • Hypersensitivity reactions	• Low interaction potential

ROFLUMILAST AND N-ACETYLCYSTEINE (ORAL)

Drugs	MOA	Safety	D/I
Roflumilast (oral)	PDE 4 inhibition, reduces inflammation	- Nausea & vomiting - Headache - Diarrhea - Psychiatric symptoms - Weight loss & decreased appetite - Abdominal pain - Contraindicated if moderate to severe hepatic impairment	Strong CYP inducers
N-Acetylcysteine (oral)	Lowers the viscosity of mucous (mucolytic)	- Stomatitis - Rhinorrhea - Nausea and vomiting	Nitroglycerin

ANTIBIOTICS – LONG-TERM MACROLIDES

Drugs	MOA	Safety	D/I
Azithromycin	Anti-inflammatory effects	- Diarrhea, nausea - QTc prolongation	- Other QTc prolonging agents - Digoxin - Warfarin

SYSTEMIC CORTICOSTEROIDS

Drugs	MOA	Safety	D/I
• Prednisone (PO) • Methylprednisolone (IV)	Decreases inflammation due to anti-inflammatory properties	• GI upset • Water/electrolyte imbalances • Cutaneous effects • Insomnia or euphoria • Increased bleeding risk and ulcers • Hypertension • Hyperglycemia & glycosuria • Cushing's Syndrome • HPA suppression • Cataracts and glaucoma • Rarely avascular necrosis of bone	• NSAIDS/COX-2 Inhibitors • Phenytoin • Rifampin • CYP3A4 inhibitors/inducers • Vaccines • Diuretics

INHALER DOSAGE FORMS

Class	Drugs	Dosage Form
SABA	Terbutaline	DPI Turbuhaler
	Salbutamol	MDI or DPI Diskus
LABA	Salmeterol	DPI Diskus
	Formoterol	DPI Turbuhaler
	Indacaterol	Breezhaler

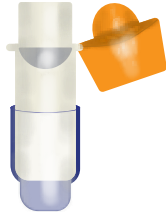
INHALER DOSAGE FORMS

Class	Drugs	Dosage Form
LAMA	Tiotropium	HandiHaler or Respimat
	Glycopyrronium	Breezhaler
	Aclidinium	DPI Genuair
	Umeclidinium	DPI Ellipta
SAMA	Ipratropium	MDI
SABA/SAMA	Salbutamol/Ipratropium	Respimat
ICS	Beclomethasone Dipropionate	MDI
	Budesonide	DPI Turbuhaler
	Ciclesonide	MDI
	Fluticasone Propionate	MDI or DPI Diskus
	Mometasone furoate	DPI Twisthaler

INHALER DOSAGE FORMS

Class	Drugs	Dosage Form
ICS/LABA	Fluticasone/Salmeterol	MDI or DPI Diskus
	Budesonide/Formoterol	DPI Turbuhaler
	Mometasone Furoate/Formoterol	MDI
	Mometasone /Indacaterol	Breezhaler
	Fluticasone/Vilanterol	DPI Ellipta
LABA/LAMA	Olodaterol/Tiotropium	Respimat
	Vilanterol/Umeclidinium	DPI Ellipta
	Indacaterol/Glycopyrronium	Breezhaler
	Formoterol fumarate dihydrate/Aclidinium	Genuair
ICS/LAMA/LABA	Fluticasone/Umeclidinium/Vilanterol	DPI Ellipta
	Budesonide/Glycopyrronium/Formoterol	MDI Aerosphere
	Mometasone/Glycopyrronium/Indacaterol	Breezehaler

DEVICE COMPARISON

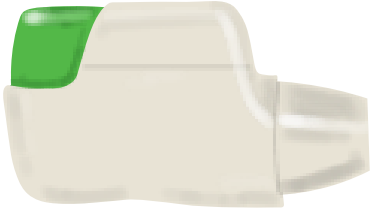
Device Type	PROS	CONS	Example
MDI	Can be used in all ages	- Needs priming - Some patients (e.g. young children) may require a spacer device (e.g. aerochamber)	
Diskus*	Dose counter	Once removed from package; short expiry date	
Respimat	Dose counter - Mist released over a few seconds → may improve technique	Needs priming	



*Requires a forceful and sharp inhalation to get full dose

Tipping over the device before inhaling it can remove the medication

DEVICE COMPARISON

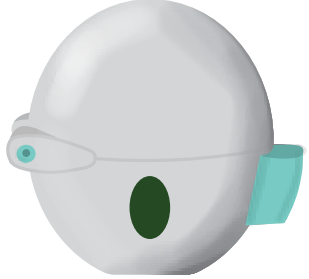
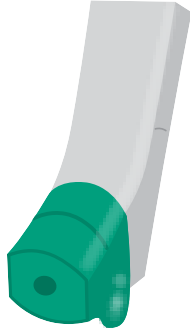
Device Type	PROS	CONS	Example
Turbuhaler*	• Dose counter • Dose not lost if device twisted a few times	• Desiccant can be heard when empty; pts may think there are doses left • Needs to be primed	
Genuair*	• Easy to use • Dose counter • Visual/audible feedback when dose taken correctly	• May taste bitter	
Ellipta*	• Easy to use • Dose counter • Large numbers used in dose counter	• No feedback on if dose taken correctly • Short expiry once opened	



*Requires a forceful and sharp inhalation to get full dose

Tipping over the device before inhaling it can remove the medication

DEVICE COMPARISON

Device Type	PROS	CONS	Example
Handihaler Breezhaler Aerolizer	- Each capsule = 1 dose - Low effort needed when inspiring - Rattling is heard when capsule contents have been properly inhaled	- Multiple steps - Capsules come in foil; may be hard to open	
Respiclick	- One step prep Dose counter	- Looks like an MDI but a DPI - Removal of cap will load dose	



*Requires a forceful and sharp inhalation to get full dose

Tipping over the device before inhaling it can remove the medication

REFERENCES

1. COPD management. *Can Respir J* Vol 14 Suppl B Sept 2007
2. Chronic Obstructive Pulmonary Disease. In: RxTx. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/new/documents/CHAPTER/en/c0040>
3. Criner GJ, Bourbeau J, Diekemper RL, et al. Executive summary: Prevention of acute exacerbation of copd: american college of chest physicians and Canadian thoracic society guideline. *Chest*. 2015;147(4):883-893. doi:10.1378/chest.14-1677.
4. Marciniuk DD, Goodridge D, Hernandez P, et al. Managing dyspnea in patients with advanced chronic obstructive pulmonary disease: a Canadian Thoracic Society clinical practice guideline. *Can Respir J*. 2011;18(2):69-78.
5. Asthma In Adults. In: RxTx. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/new/>
6. Product monographs for each referenced product

CHANGE LOG

July 2025

- Minor formatting and grammatical changes made
- Slide 5: Removed duplicate information

October 2025

- Slide 12: Removed oral erythromycin as it is no longer available on the Canadian market

March 2026

- Slide 2: Added NAPRA competencies slide
- Minor formatting and grammatical edits